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A PALAEOMAGNETIC RECONNAISSANCE OF NORTHEASTERN BALUCHISTAN, PAKISTAN

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Abstract. A total of 560 samples from three areas in northeastern Baluchistan (the southern Sulaiman Range, the central Loralai Range, and the northern Kirthar Range) were analyzed using thermal demagnetization techniques. Thirteen formations of late Palaeozoic to early Tertiary age were studied palaeomagnetically. Inclinations of the obtained results show a general affinity with the Indian apparent polar wander path. Deviating declinations from the Loralai Range indicate a clockwise rotation over 50° with respect to the Indian shield. Secondary magnetization components probably of late Palaeocene to early Eocene age were consistently present in the samples from the Kirthar Range and the Sulaiman Range but were not observed in samples from the Loralai Range. Acquisition of these components is attributed to crustal upwarping during the Palaeocene, which is tentatively related to initial collision of continental Indo-Pakistan with an island arc off south central Asia. The Baluchistan data support recent palaeomagnetic results from the Indus-Tsangpo suture zone in Ladakh (NW Himalaya), which indicate that such an initial collision took place at very low northern palaeolatitudes.

Introduction

It is well established from seafloor spreading data and palaeomagnetic studies that the Indo-Pakistani subcontinent has drifted north more than 5000 km following the late Mesozoic breakup of Gondwanaland. Considerable progress has been made during the last decade in unravelling the seafloor spreading record of the Indian Ocean (cf. McKenzie and Sclater, 1971; Fischer et al., 1971; Sclater and Fischer, 1974; Markl, 1974; Molnar and Tapponnier, 1975; Schlich, 1975; Larson, 1975, 1977; Johnson et al., 1976; Luyendyk and Rennick, 1977; Norton and Molnar, 1977; Weissel et al., 1977; Larson et al., 1978; Powell, 1979; Norton and Sclater, 1979). However, reconstruction of the complex fragmentation process and tracing of Indo-Pakistan's drift relative to Australia-Antarctica is hindered by (1) the obvious difficulty in detailing seafloor spreading during the Cretaceous normal polarity interval (late Aptian to early Campanian (Irving and Pullaiah, 1976; Lowrie and Alvarez, 1977; VandenBerg et al., 1978; Channell et al., 1979) and (2) failure to identify in the western Indian Ocean a magnetic anomaly pattern between 15 and 20 m.y. B.P. (late Eocene). Moreover, the reliability of the early Tertiary trajectory of the Australian apparent polar wander path (APWP) (McElhinny et al., 1974), used in a number of studies for absolute positioning of the Indian plate, has been disputed recently (Peirce, 1978; Klootwijk, 1979a; Klootwijk and Peirce, 1979). These ambiguities have so far precluded agreement over major questions like the prebreakup position of the Indo-Pakistani subcontinent with respect to Africa (Du Toit, 1937; Smith and Hallam, 1970; Tarling, 1972; Barron et al., 1978; Klootwijk, 1979a; Powell, 1979; Powell et al., 1980) and the timing, palaeoposition, and evolution of the India-Asia collision.

Estimates for the time of the India-Asia collision range from pre-middle Eocene, probably about 55 m.y. ago (Powell and Conaghan, 1973, 1975; Johnson et al., 1976; Powell, 1979), to the Eocene-Oligocene time boundary (Dewey and Burke, 1973; Molnar and Tapponnier, 1975). A more detailed record of Indo-Pakistan's northward drift and deformation of its northern boundary during collision with Asia is clearly desirable. The late Palaeozoic to early Tertiary sedimentary succession from the Pakistani fold belt (Kirthar Range, Loralai Range, and Sulaiman Range) and the Salt Range offer promising prospects for indicating details of India's drift. The rather undisturbed and largely miogeosynclinal sedimentation record of these mountain ranges is more complete than contemporaneous successions on the Indian mainland (i.e., the Cutch and Jaisalmer successions) and is far more accessible and structurally less complex than successions of similar age in the Himalaya. Moreover the stratigraphy and structure of a large part of the Pakistani fold belt are well known from geological mapping at a scale of 1:250,000 (Hunting Survey, 1960).

An allochthonous nature for the Sulaiman lobe (Loralai Range-Sulaiman Range) and the (Trans-

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Indus) Salt Range-Potwar Plateau unit has recently been proposed (Templeton, 1973; Powell, 1979; Sarwar and De Jong, 1979; Seeber and Armbruster, 1979). Establishment of palaeomagnetic constraints on the tectonic evolution of parts of the Pakistani fold belt was therefore undertaken as another objective of the present study.

We present data from a palaeomagnetic reconnaissance of three late Palaeozoic-early Tertiary successions in northeastern Baluchistan: (1) in the northern extreme of the Kirthar Range, (2) in the central Loralai Range, and (3) in the southern Sulaiman Range.

Regional Geological Setting

The Pakistani fold belt comprises a system of mountain ranges which extend from the northwestern Himalaya southward to the Arabian Sea (Figure 1). This belt is bordered to the west by the north-south trending Chaman transform fault zone (Lawrence and Yeats, 1979), which represents the presently active northwestern border of the Indian plate. To the west of this megashear the Tertiary sedimentary pile of Makran represents a wide arc-trench sequence which has been deformed by subduction of the oceanic front of the Arabian plate underneath the Dahst-I-Margo block (Farhoudi and Karig, 1977; White, 1979; Jacob and Quittmeyer, 1979; Arthurton et al., 1979).

To the east of the Chaman megashear, four lithologically and structurally different zones can be recognized, eastward toward the Indian shield :

1. There is a zone of Tertiary flysch, largely of Oligocene and Miocene age, which resembles the north Makran sediments. This zone is present over the length of the megashear, but its width increases from 30 km in the south to more than 200 km in the Zhob-Kakar zone.

2. A discontinuous zone of ultramafic complexes (Las Bela, Zhob Valley, Waziristan) and associated mélanges (Hunting Survey, 1960, Gansser, 1966, 1979; Asrarullah et al., 1979; Abbas and Ahmad, 1979; De Jong and Subhani, 1979) exists which was obducted onto the Indo-Pakistani subcontinent during the Palaeocene or perhaps the early Eocene (Allemann, 1979). Their present outline is up to 200 km east from the Chaman megashear.

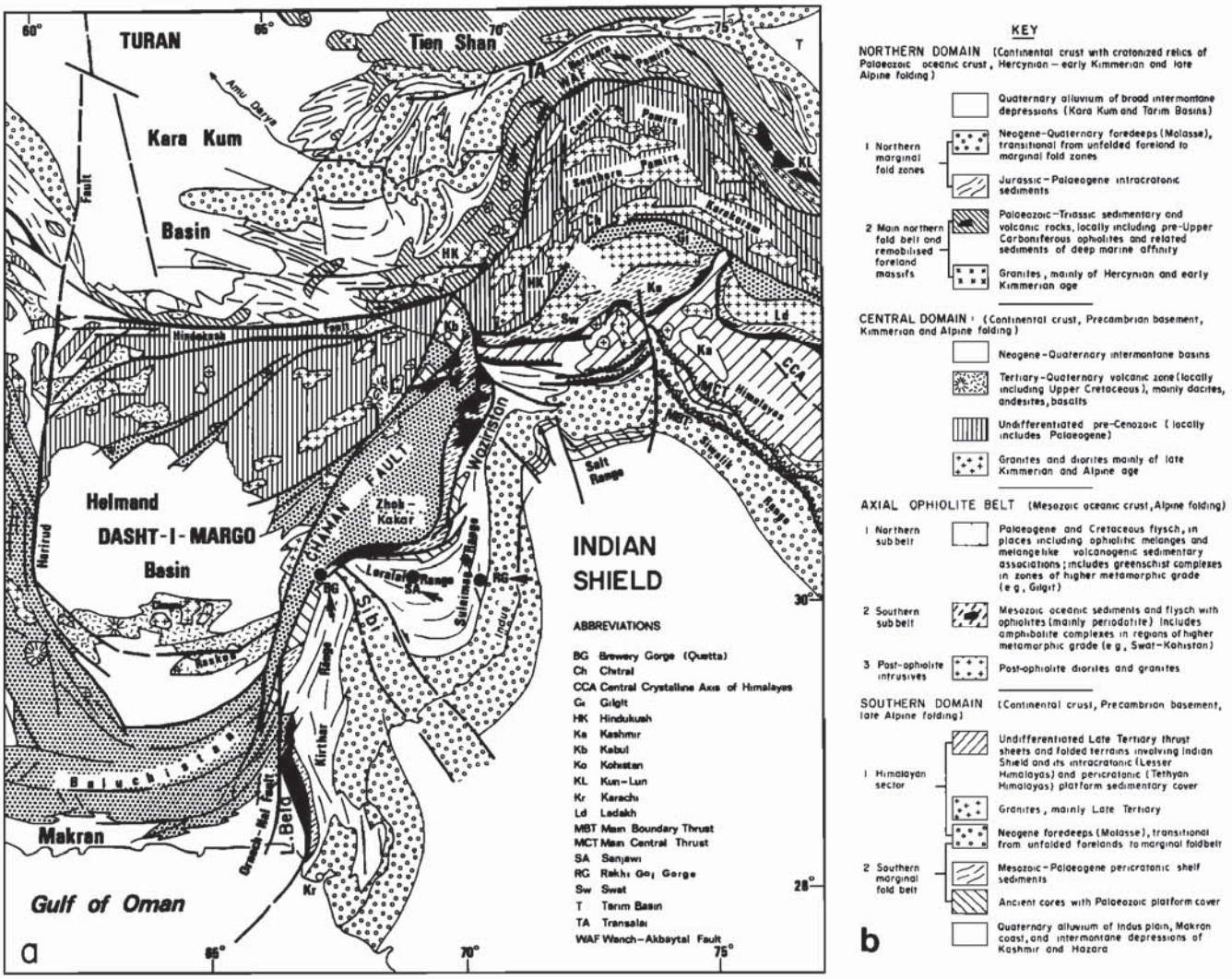


Fig. 1. (a) Regional tectonic zonation of the study area, slightly adapted from Stöcklin (1977). Dots with arrows indicate the sample localities. (b) May key to Figure 1a.

3. Prominent in the actual fold belt is the thick succession of Jurassic and subordinate Permo-Carboniferous to Triassic limestones, shales and dolomites. Overlying Cretaceous shales and carbonates become of a more pelagic character toward the interior of the fold belt. Toward the shield the Cretaceous succession becomes more marly and sandy. Deposition was influenced by positive tectonic structures like the Sanjawi Arch (Hunting Survey, 1960), an anticlinal feature extending more than 200 km from the region north of Quetta through Sanjawi (Figure 1, SA) eastward. This arch originated during the Cretaceous and became eroded during the Palaeocene. At the end of the Palaeocene, erosion of Upper Cretaceous and Palaeocene sediments occurred over considerable regions of the outer part of the fold belt. The Eocene was transgressive, but influx of terrigenous components became more abundant during the Oligocene. This heralded the onset of the Mio-Pliocene molasse sedimentation.

4. The Mio-Pliocene molasse sedimentation, which borders the fold belt in a wide zone to the east, is particularly prominent in the Sibi Trough (i.e., the apex of the Quetta Syntaxis), which became filled with a molasse sequence approximately 7 km thick.

The Pakistani fold belt forms a typical foreland fold-and-thrust belt with tectonic transport toward the Indian shield. Characteristic features are a series of fold festoons and intervening syntaxes, most notably being the Loralai-Sulaiman festoon and the Quetta Syntaxis (Hunting Survey, 1960; Voskresenskiy et al., 1968; Templeton, 1973; Andrieux and Brunel, 1977; Powell, 1979; Sarwar and De Jong, 1979). Juratype deformation and seismic evidence (Nowroozi, 1972; Menke and Jacob, 1976; Rowlands, 1978; Quittmeyer et al., 1979) suggest that this festoon-reentrant structure resulted from décollement of supracrustal rocks over the basement of the Indian shield.

The major tectogenic phase of the Pakistani fold belt is very young, Plio-Pleistocene in the more interior part and Pleistocene-Recent toward the Indian shield. This décollement may be presently continuing as shown by high seismic activity to the south and southeast of the Sulaiman lobe (Menke and Jacob, 1976; Seeber and Armbruster, 1979; Quittmeyer et al., 1979).

Sampling

For this reconnaissance study, sampling was restricted for three areas in the Quetta-Loralai-Dhera Ghazi Khan region (Figure 1), which were chosen because their structure represents the main trend of the Quetta Syntaxis and the Sulaiman festoon. The sampled areas were (1) the Brewery gorge, directly west of Quetta (Figure 1, BG) near the northern end of the north-south trending Kirthar Range, (2) the Sanjawi area (Figure 1, SA) in the east-west trending Loralai Range, and (3) the Rakhi Gaj gorge (Figure 1, RG) east of Fort Munro in the southern part of the north-south trending Sulaiman Range.

The Rakhi Gaj gorge and the Sanjawi area were sampled during December 1976, using a portable field drill. The Brewery gorge was sampled in July-August 1977. At the latter locality, block samples were collected, from which cores were drilled with a drill press at the premises of the Geological Survey of Pakistan at Quetta. All samples were oriented in the field with both a magnetic and a solar compass, using a device described by Embleton and Edwards (1973).

A brief description of the main characteristics of the sampled formations and sampling details is now presented. For further stratigraphic detail, the reader is referred to the Hunting Survey (1960) report and to Shah (1977), and to Allemann (1979) for the Brewery gorge.

Quetta Area

A Middle Jurassic to upper Palaeocene succession was sampled in the monoclinal northwest dipping section exposed in the Brewery gorge (66°58'E, 30°10'N), west of Quetta. The older Alozai Formation was sampled farther south in the Ziarat Nala (66°51'E, 30°1'N), and the Spintangi Limestone in some isolated hillocks northeast of Shaikh Wasil (66°40'E, 29°55'N).

Alozai Formation. Thirty-six block samples were taken in well-bedded grey limestones from three sites (PZAL01-36, N200°E/60°W), each about 100 m stratigraphically apart and starting about 150 m above the reverse fault contact with the Chiltan Limestone in the entrance of the Ziarat Nala. The age of the Alozai Formation is not well defined and may range from Permo-Carboniferous to early Liassic.

Chiltan Limestone. Thirty block samples (PMCH01-30, N225°-245°E/40°-49°NW) were taken in dark grey oolitic and massive limestones over a stratigraphic interval of about 15 m in the top part of the formation, which is dated as Oxfordian to lower Kimmeridgian (Allemann, 1979).

Parh Group. Fifty-one block samples (PMGU01-29, PMPA01-22, N220°-230°E/32°-50°NW) were taken over a stratigraphic interval of about 120 m through a multicolored pelagic limestone section, which ranges from Aptian-Albian to Campanian-Santonian (Allemann, 1979). The lower part of the Parh Group (Hunting Survey, 1960) has been redefined (Shah, 1977) as the Goru Formation. In the Brewery gorge its age ranges from Aptian-Albian to Turonian-Coniacian. Samples PMGU01-29 were taken through this succession. The upper part of the Parh Group (Hunting Survey, 1960) is redefined as Parh Limestone (Shah, 1977). Samples PMPA01-22 were taken from this section, which ranges from Turonian-Coniacian to Santonian-Campanian.

Brewery Limestone. A stratigraphic succession of about 40 m of massive dark grey bioclastic limestone succeeds the Parh Limestone in the Brewery gorge, probably with an angular unconformity. Twenty-six block samples (PFMU01-26, N220°-225°E/40°-45°NW) were taken over a stratigraphic interval of about 20 m through the lower part of this formation (upper Maastrichtian) (Allemann, 1979). Another 13 block samples (PMDU01-13, N220°E/42°W) were taken through the top 3 m of the formation, dated as late Palaeocene (Allemann, 1979).

Spintangi Limestone. Thirty block samples (PSWS01-24, N240°-280°E/27°NW, PSWS25-30, N330°-340°E/67°E) were taken in grey to blue-grey massive limestones (lower to middle Eocene), forming isolated hillocks northeast to Shaikh Wasil.

Sanjawi Area

In the central Loralai Range, Middle Jurassic and Cretaceous carbonates were sampled on the northern limb of an east-west aligned anticlinal structure plunging slightly westward toward Old Sanjawi (68°22'E, 30°16'N). Lower Tertiary carbonates were sampled in the southern limb.

Loralai Limestone. Seventy-five cores (PLL001-75, N250°-260°E/33°-36°N) were drilled over a stratigraphic interval of about 45 m in the exposed top of the formation, directly south of the Loralai-Sanjawi road. The samples were taken from algal limestones, thinly bedded at the top but more massive toward the basal part (upper Liassic) of the formation. An upper age limit for this formation is not firmly established, but the sampled beds are most probably of Middle Jurassic age. The Loralai Limestone can be correlated with the Chiltan Limestone, sampled near Quetta.

Parh Group. Sampled directly north of Old Sanjawi, a total of 51 cores (PLGU01-51, N235°E/61°NW) were collected over a stratigraphic interval of 70 m in green sublithographic limestones from the top part of the Goru Formation. Six cores (PLPA52-57, N245°E/63°NW) were taken over about 6 m stratigraphically in a conspicuous purple flaggy and sublithographic limestone unit at the base of the Parh Limestone. Slumping is very distinct in the higher part of the Parh Limestone, and another 15 cores (PLPA58-72, N235°-250°E/57°-62°NW) were taken over about 6 m stratigraphically in the least disturbed part of this zone, which is about 20 m above the base of the Parh Limestone.

Sanjawi Limestone. Sampled east of Sanjawi in the road cut along the Sanjawi-Duki road. The top and basal part of this formation are made up of nodular limestones. Sixty-seven cores (PLSA01-67, N80°-90°E/60°S) were taken over about 40 m stratigraphically from well-bedded limestones in the middle part of this formation (upper Palaeocene-lower Eocene).

Rakhi Gaj Gorge

An upper Cretaceous to lower Tertiary section was sampled in the gorge (70°5'E, 29°58'N) along the Fort Munro-Dera Ghazi Khan road in the eastern limb of an anticlinal structure in the external part of the southern Sulaiman Range.

Mughal Kot Formation. This formation was sampled in the core of the anticlinal structure. Forty-one cores (PRMK01-41, N340°-355°E/14°E) were taken over 20 m stratigraphically in dark grey argillaceous claystones in the basal part of the formation. The age of this formation ranges from Campanian to Maastrichtian (Shah, 1977).

Fort Munro Formation. A total of 42 cores (PRFM01-42, N0°-15°E/22°E) were drilled over 15 m stratigraphically in dark grey massive limestones in the middle part of the formation, which is dated as late Campanian to early Maastrichtian (Shah, 1977).

Kharo Formation. A total of 37 cores (PRGB01-37, N5°-15°E/36°E) were drilled in brown-colored to locally ferrugineous, thick-banked sandstones in the top 25 m of this formation, dated as Danian (Shah, 1977).

Habib Rahi (Zam) Limestone. A total of 37 cores (PRHR01-37, N20°-30°E/64°E) were drilled over a stratigraphic interval of 20 m in thin-bedded argillaceous limestones in the middle part of the Habib Rahi Limestone (early middle Eocene (Shah, 1977) of the Kirthar Formation.

Method of Analysis

Generally, two specimens, 2.5 cm in diameter and 2.2. in length, were sliced from each sample. One specimen from every second or third sample throughout a sampled succession was subjected to a pilot progressive thermal demagnetization treatment in 12-18 steps from room temperature till the magnetic intensity became too weak for directional measurements (i.e., total moment less than 3.10^{-2} mAm^{-1}). The remainder of the specimens were subjected to partial progressive thermal demagnetization in 8-10 steps over a temperature interval selected from the pilot studies. All heatings were done in a furnace (McElhinny et al., 1971), in which the magnetic field remained less than 5 nT during the course of the heating-cooling cycle (approximately 2 hours) with the aid of an automatic field cancellation system. All specimens were degassed and flushed with argon gas prior to the heating sequence. All heatings were done in argon gas to reduce possible oxidation effects. Care was taken to keep the specimens magnetically shielded to avoid excessive builup of viscous remanent magnetization (VRM) which developed after heating at the higher temperatures. Changes in magnetic susceptibility during the course of the progressive thermal demagnetization treatment were monitored with a Digico bulk susceptibility unit. All natural remanent magnetization (NRM) measurements were made on a superconducting rock magnetometer (Sct, noise level less than 2.10^{-2} mAm^{-1}) interfaced with a Digico minicomputer for immediate data reduction.

Several pilot specimens from each formation were subjected to progressive af demagnetization along the three axes (Zijderveld, 1967, 1975) in 12-20 steps from 2.5 mT up to 210 mT peak field. Separation of the various magnetic components was less successful than that with the thermal treatment, and further af demagnetization was therefore not pursued.

Directions of the characteristic and secondary components were identified from orthogonal projection figures and were computed with a principal component analysis program adapted after Kirschvink (1980).

Individual specimens from the same sample (generally, no more than two) occasionally displayed variable magnetic properties and appreciable differences in direction upon thermal demagnetization. For this reason and also because of the sectionwise rather than sitewise collection of samples, individual specimen directions were given unit weight in the calculation of formation mean directions. The nonpelagic carbonates, in particular, showed often overlapping blocking temperature spectra for the various magnetic components. Simultaneous elimination during demagnetization frequently prevented determination of individual component directions. For this reason and also because of the predominance of a recent field component, we were unable to determine characteristic or secondary

TABLE 1A SUMMARY OF PALAEOMAGNETIC RESULTS NORTHEASTERN BALUCHISTAN

FORMATION	AGE	INITIAL NRM DATA 1) Samples	2) Intensity	3) Comp.	AFTER THERMAL DEMAGNETIZATION AND BEDDING CORRECTION MEAN SPECIMEN DIRECTION Decl.	Incl.	K	α_{95}	4) N	S. POLE POSITION Lat.	Long.	dp.	dm.	SAMPLE ROTATION
A) RAKHI GAJ GORGE (70°5'E 29°58'N)														
HABIB RAHI	M. EOCENE	37(68)	89-427	pr?	21.6°	+19.0°	17.0	13.8°	8	61.5°S	20.7°E	7.5°	14.4°	PRHR
				sc.5)	176.4°	-52.4°	36.8	9.3°	8	85.7°S	154.2°E	8.8°	12.8°	"
KHADRO FM.	DANIAN	37(51)	8.5-79240	sc?	169.3°	- 5.6°	38.6	3.2°	51	61.0°S	92.7°E	1.6°	3.2°	PRGB
FORT MUNRO FM.	MAASTRICHTIAN	42(58)	200-1419	pr.	139.0°	+32.7°	12.6	27.0°	4	28.0°S	115.2°E	17.2°	30.4°	PRFM
				sc.	25.4°	+23.9°	21.5	5.4°	34	60.7°S	11.2°E	3.1°	5.8°	"
MUGHAL KOT FM.	CAMPANIAN - MAASTRICHTIAN	41(72)	236-1725	pr.	305.2°	-58.1°	50.3	2.7°	56	4.4°S	109.9°E	3.0°	4.0	PRMK
				sc	20.6°	+ 8.3°	21.8	4.2°	55	57.7°S	29.1°E	2.1°	4.2°	"
B) SANJAWI (68°22'E 30°16'N)														
SANJAWI LST.	PALAEOCENE-EOCENE	67(130)	8-192	pr.	22.0°	+15.0°	13.5	14.5°	9	59.4°S	21.8°E	7.7	14.9°	PLSA
PARH LST.	CONIACIAN TO SANTONIAN/CAMPANIAN	21(28)	169-2188	pr.	7.1°	-45.6°	24.2	12.5°	7	32.2°S	61.0°E	10.2°	15.9°	PLPA
				sl.	333.5°	-56.4	25.6	8.8°	12	-	-	-	-	"
GORU FM.	APTIAN/ALBIAN TO CONIACIAN	51(79)	49-2882	pr.	358.7°	-40.2°	34.4°	3.0°	69	36.8°S	70.1°E	2.2°	3.6°	PLGU
GORU FM.-PARH LST. COMBINED				pr.	359.3°	-40.6°	32.5	2.9°	76	36.5°S	69.3°E	2.1°	3.5°	
LORALAI LST.	M. JURASSIC	75(112)	10-2274	pr.	338.2°	-42.9°	8.8	8.0°	41	31.0°S	91.6°E	6.1°	9.9°	PLLO
C1) BREWERY GORGE (66°58'E 30°10'N)														
BREWERY LST. (top)	base UPPER PALAEOCENE	13(35)	16-434	pr.	331.6°	+ 8.4°	8.5	22.0°	7	52.7°S	118.5°E	11.1°	22.1°	PMDU
BREWERY LST. (base)	UPPER MAASTRICHTIAN	26(93)	38-760	sc.	144.1°	-28.8°	44.1	3.3°	44	54.0°S	140.9°E	2.0°	3.6°	PFMU
PARH LST.	CONIACIAN TO SANTONIAN/CAMPANIAN	22(89)	82-339	pr.	302.1°	-42.2°	33.0	4.2°	38	12.2°S	119.1°E	3.2°	5.1°	PMPA
GORU FM.	APTIAN/ALBIAN TO CONIACIAN	29(73)	126-4285	pr.	310.0°	-42.5°	16.3	5.3°	48	17.2°S	113.7°E	4.0°	6.5°	PMGU
				sc.	140.7°	-35.1°	34.1	7.8°	11	52.3°S	147.4°E	5.1°	9.0°	"
GORU FM.-PARH LST. COMBINED				pr.	307.0°	-43.6°	16.5	3.9°	86	14.7°S	115.2°E	3.0°	4.8°	
CHILTAN LST.	OXFORDIAN TO LOWER KIMMERIDGIAN	30(123)	10-119	pr.	103.8°	+38.0°	16.9	10.4°	13	0.5°S	131.7°E	7.3°	12.3°	PMCH
C2) ZIARAT NALA (66°51'E 30°1'N)														
ALOZAI FM.	PERMO-CARBONIFEROUS?	56(153)	218-4039	pr.	276.5°	-66.0°	6.1	12.1°	28	18.1°N	111.0°E	16.1°	19.7°	PZAL
				sc.	147.0°	-25.4°	15.2	7.7°	25	55.2°S	135.3°E	4.5°	8.3°	"
C3) SHAIKH WASIL (66°40'E 29°55'N)														
SPINTANGI LST.	L.-M. EOCENE	30(113)	62-1154	??	138.0	+50.2°	12.5	11.3°	15	17.0°S	103.8°E	10.1°	15.1°	PSMS

TABLE CAPTIONS

1) Number of specimens between parentheses
2) Intensity in units of 10^{-3} mAm^{-1}
3) pr. = primary magnetization component
   sc. = secondary magnetization component
   sl. = slumped sequence.
4) Numbers of specimens wherein the component concerned could be isolated.
5) Probably of late Tertiary age.

magnetic component directions in an appreciable number of specimens (Table 1).

Results

A magnetic component of recent origin dominated the NRM of the carbonate specimens, particularly those of nonpelagic origin, with NRM directions streaking toward the present field direction (Figures 2, 3, 4a, 4b and 4c). This component was removed in nearly all specimens after heating up to 150°C to 200°C. Another secondary component (Quetta area and Rakhi Gaj gorge specimens only) and a characteristic component (all three areas) were removed at

TABLE 1B SUMMARY OF PALAEOMAGNETIC RESULTS FROM THE LORALAI RANGE - CORRECTED FOR INFERRED ROTATION

FORMATION	COMP.	ORIGINAL Decl.	Incl.	ROTATION	ANGLE	ROTATED Decl.	Incl.	ROTATED S.POLE POSITION Lat.	Long.
SANJAWI									
SANJAWI LST.	pr.	22.0°	+15°	50°	clockwise	332.0°	+15.0°	55.3°S	123.5°E
PARH LST.	pr.	7.1°	-45.6°	50°	clockwise	317.1°	-45.6°	19.5°S	108.5°E
GORU FM.	pr.	358.7°	-40.2°	50°	clockwise	308.7°	-40.2°	17.5°S	117.5°E
GORU FM. - PARH LST. COMBINED	pr.	359.3°	-40.6°	50°	clockwise	309.3°	-40.6°	17.7°S	116.8°E
LORALAI LST.	pr.	338.2°	-42.9°	50°	clockwise	288.2°	-42.9°	1.9°S	128.1°E

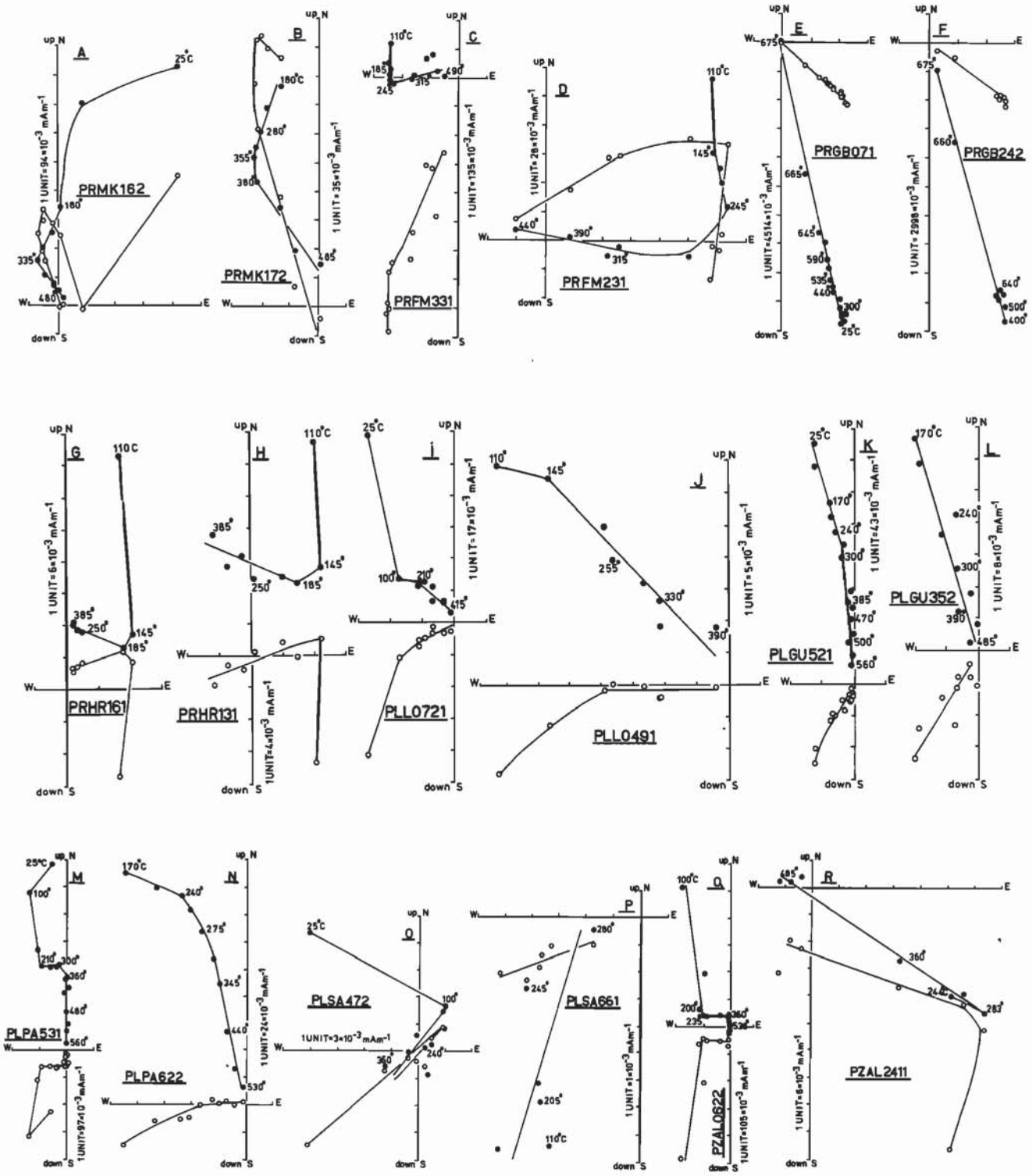


Fig. 2. Thermal demagnetization diagrams of representatives specimens. The points denote successive positions-in orthogonal projection-of the endpoints of the resultant magnetization vector during progressive demagnetization. Solid circles denote projections on the vertical east-west plane, and open circles denote projections on the horizontal plane.

higher temperatures. As will be discussed in the next section, this secondary component is most probably of early Tertiary age, and the characterisitic component represents for nearly all formations the primary magnetization.

The magnetic susceptibility generally increased rapidly after thermal treatment above 450°C to 500°C, and an increase of VRM generally prevented meaningful remanence measurement after treatment above 600°C (Figure 3b).

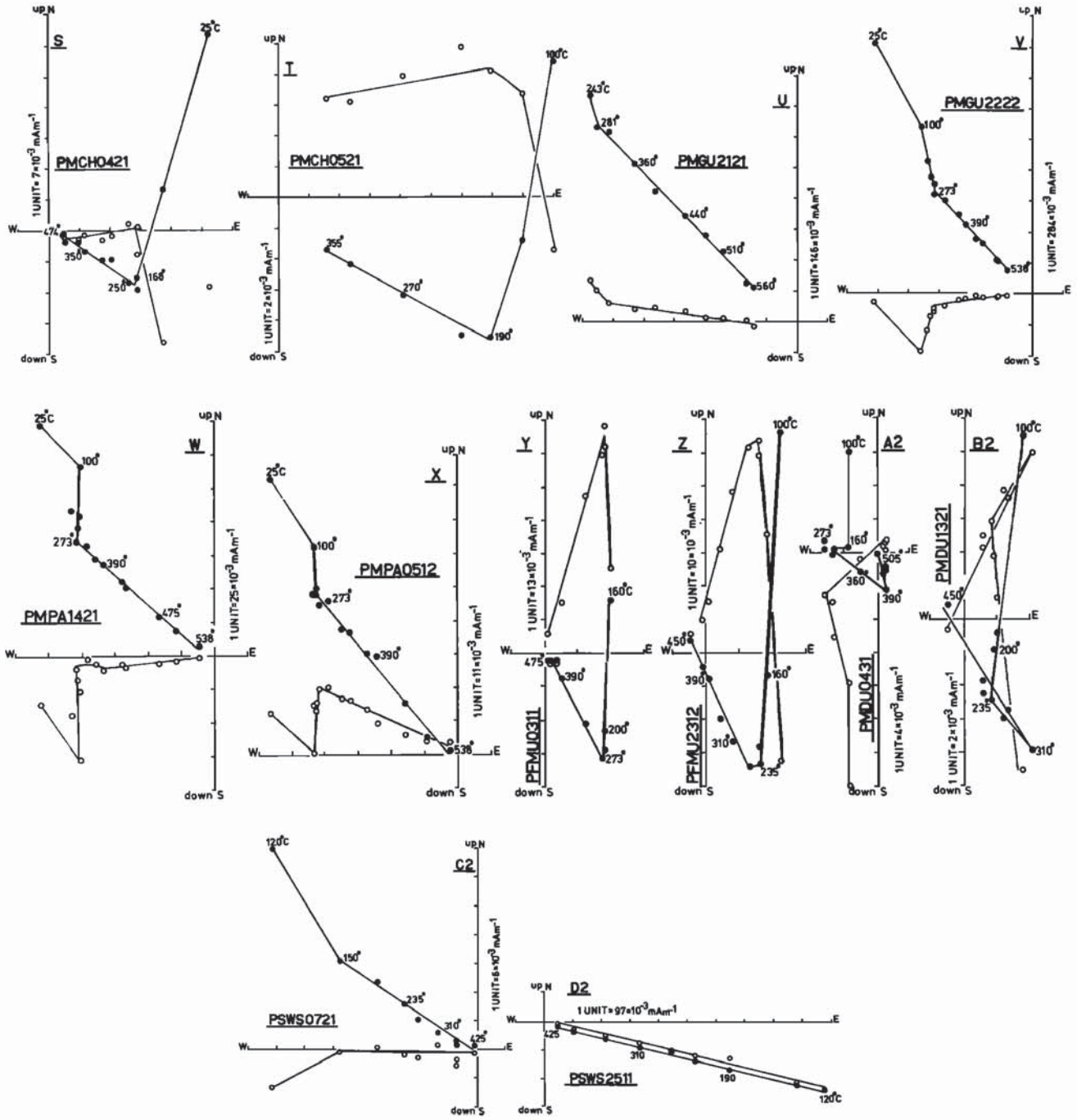


Fig. 2. (continued)

Quetta Area

Alozai Formation. Four magnetic components could be identified. A recent field component, constituting up to 80% of the NRM, was eliminated at 250°C (Figures 2q and 3a). In addition, another recent field component of minor intensity only became eliminated between 350°C and 550°C (Figure 2q). A characteristic component of normal polarity and of rather large spread in direction (Figure 4h2 and Table 1) was eliminated in the two lower sections between 250°C and 400°C (Figures 2q and 2r). In the stratigraphically highest section (PZAL25-36) a secondary component of reversed polarity was removed between 250°C and 450°C (Figure 5c2).

Chiltan Limestone. Initial intensity of magnetization was very low (10^{-1} to 10^{-2} mAm^{-1}). A recent field component, constituting up to 70% of the NRM, was generally eliminated at 250°C (Figures 2s and 2t), but in some specimens this was the only component identified during thermal treatment up to 500°C. A characteristic component of reversed and occasionally normal polarity was being removed between 250°C and 500°C (Figures 2s and 2t). Most specimens, however, showed simultaneous elimination of the characteristic and the recent field component,

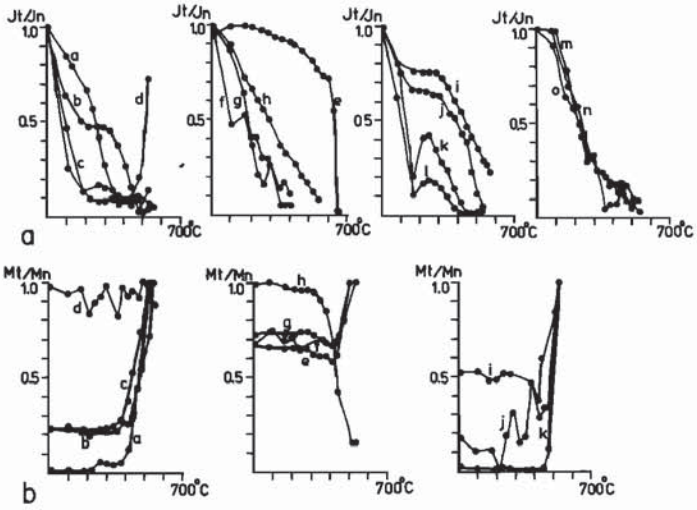


Fig. 3. (a) Normalized intensity decay curves of remanent magnetization during thermal treatment of some representative specimens. Legend: a, Goru Formation (Fm.) (Sanjawi), PLGU051; b, Mughal Kot Fm., PRMK291; c, Fort Munro Fm., PRFM371; d, Habib Rahi Limestone (Lst.), PRHR 141; e, Khadro Fm., PRGM141; f, Loralai Lst., PLLO017; g, Sanjawi Lst., PLSA292; h, Parh Lst. (Sanjawi), PLPA521; i, Goru Fm. (Brewery gorge), PMGU1511; j, Parh Lst. (Brewery gorge), PMPA0512; k, Brewery Lst. (base), PFMU1421; l, Alozai Fm., PZAL3631; m, Chiltan Lst., PMCH2321; n, Spintangi Lst., PSWS0711; and o, Brewery Lst. (top), PMDU0311. (b) Normalized intensity decay curves of induced magnetization of some representative specimens. Legend: a, Habib Rahi Lst., PRHR171; b, Mughal Kot Fm, PRMK011; c, Goru Fm. (Brewery gorge), PLGU151; d, Goru Fm. (Sanjawi), PLGU021; e, Goru Fm. (Brewery gorge), PMGU2731; f, Parh Lst. (Brewery gorge), PMPA0311; g, Alozai Fm., PZAL1021; h, Alozai Fm., PZAL0721; i, Brewery Lst. (base), PFMU 0311; j, Brewery Lst. (top) PMDU0122; and k, Spintangi Lst., PSWS2012.

which prevented determination of the direction of the characteristic component.

Goru Formation-Parh Limestone. These are undoubtedly the most suitable rocks palaeomagnetically from the Quetta area. A recent field component of minor intensity (Figures 2v-2x and 3a) became eliminated at 250°C. A well-determined characteristic component of normal polarity only was broken down between 275°C and 560°C (Figures 2u-2x, 3a, 4i, and 4j). In a number of specimens from mainly the Goru Formation, a secondary component of reversed polarity (Figure 5e) was removed between 220°C and 350°C. No consistent shift in direction of the characteristic component was observed over the sampled stratigraphic interval (Figures 4i and 4j).

Brewery Limestone (Basal Part). The magnetization of these rocks consisted of a recent field component and a secondary component of reversed polarity. The recent field component of large intensity (70-90% of NRM) was eliminated at 250°C, whereas the secondary component was broken down between 250°C and temperatures as high as 540°C (Figures 2y and 2z). No characteristic component could be identified.

Brewery Limestone (Top Part). A recent field component was very dominant in these samples (Figures 2a2, 2b2, and 3a) and was in most specimens the only component which could be identified during heating up to 540°C. A characterisitic component of both reversed and normal polarity could be determined in only a few specimens between 220°C and 500°C (Figures 2a2 and 4b2).

Spintangi Limestone. A dominant recent field component which constituted more than 90% of the NRM (Figure 3a) became eliminated largely at 150°C. A characteristic component of low intensity and of both normal and reversed polarity was broken down between 150° and 500° (Figures 2c2 and 2d2).

Sanjawi Area

Loralai Limestone. NRM intensity of most specimens was low and occasionally below the noise level of the cryogenic magnetometer, whereas the higher-intensity specimens mainly revealed a recent field component only. In a number of specimens from the upper part of the sampled succession, this recent field component contributed no more than 75% of the NRM intensity and could be removed at about 100°C to 150°C. A characteristic component of normal polarity was subsequently eliminated between 150°C and 400°C (Figures 2i, 2j, 3a and 4f).

Goru Formation-Parh Limestone. Similar to the Brewery gorge section, specimens from this succession showed very well determined palaeomagnetic directions. In the specimens from the Goru Formation a recent component or a laboratory-induced VRM was generally completely removed at 100°C to 170°C, and a characteristic component of normal polarity was removed between 150°C and 400°C (Figures 2k, 2l, 3a, and 4g). The magnetization of the overlying Parh Limestone was somewhat more complex. A recent field component was generally removed at 170°C (Figures 2m and 3a), and an intermediate component was broken down between 200°C and 300°C. Subsequently, a characteristic component of normal polarity became eliminated between 300°C and 550°C (Figures 2m and 2n). The directions of this characteristic component for the lower part of the Parh Limestone and the directly underlying Goru Formation are very similar. In contrast, the mean characteristic direction for the slumped top part of the sampled Parh Limestone shows a more westward declination by about 30°, although the inclinations are in good agreement (Table 1). Consequently, the latter directions have been discarded.

Sanjawi Limestone. These samples has a very low NRM intensity (10^{-1} to 10^{-2} mAm^{-1}), and most samples contained a recent field component only. A characteristic component of normal polarity and of very low intensity could be determined between 100°C and 400°C in a minority of the specimens only (Figures 2o and 2p).

Rakhi Gaj Gorge

Mughal Kot Formation. A laboratory-induced VRM and/or a magnetically soft recent field component of substantial intensity (up to 50% of NRM) was removed at 100°C. A secondary compo-

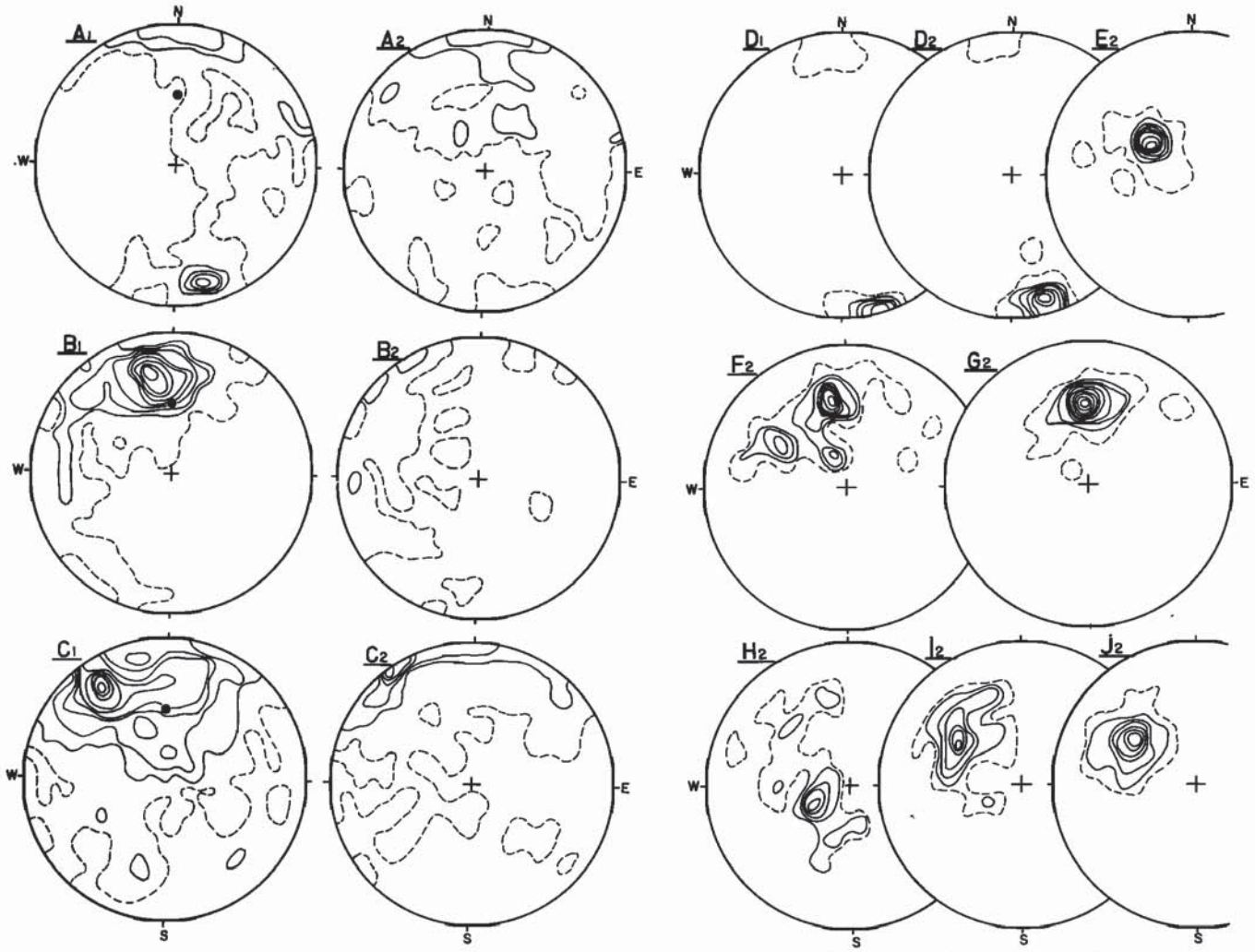


Fig. 4. Frequency distribution of bedding-corrected (not Figures 4a-4c) specimen directions in equal-area projection. A 1% counting circle was applied. Subscript 1 denotes lower hemisphere projection, and subscript 2 denotes upper hemisphere projection. Frequency intervals are indicated hereunder in parenthesis: (a) NRM specimen directions combined for the Rakhi Gaj gorge (5). (b) As above, Sanjawi area (5). (c) As above,Quetta area (5). (d) Primary direction, Khadro Fm. (5). As above, Mughal Kot Fm. (5). (f) As above, Loralai Limestone (1). (g) As above, Goru Fm.-Parh Lst. combined at Sanjawi (3). (h) As above, Alozai Fm. (1), (i) As above, Goru Fm. at Brewery gorge (2). (j) As above, Parh Lst. at Brewery gorge (2). The dots indicate the downwards pointing present field direction at the sampled localities.

nent of normal polarity was removed between 180°C and 330°C (Figures 2a, 2b and 5a), and a well-determined characteristic component of normal polarity only was eliminated between 350°C and 480°C (Figures 2a, 2b, 3a and 4e).

Fort Munro Formation. In these samples a soft viscous magnetization (up to 80% of NRM) was generally removed at 110°C. As with the Mughal Kot Formation, a secondary component of normal polarity and of appreciable intensity was removed between 100°C and 250°C to 300°C. The magnetization remaining after treatment at higher temperatures was genrally difficult to interpret apart from a few specimens in which a characteristic component of both normal and reversed polarity could be identified between 250°C and 500°C (Figures 2c and 2d) (Table 1).

Khadro Formation. This formation constitutes the only red beds studied, and they gave the most precise pole obtained in the present reconnaissance study. Unfortunately, this pole probably does not represent the primary magnetization, as will be discussed. A weak recent field component (less than 20% of NRM) was generally removed at 180°C. At higher temperatures, only a characteristic component became removed, very gradually up to 650°C, followed by a very fast decay over a discrete blocking temperature range between 650°C and 675°C (Figures 2e, 2f, and 3e). This component is dominantly of reversed polarity, but some normal polarity directions were found in the top part of the sampled section.

Habib Rahi Limestone. A dominant recent field component (up to 90% of NRM) became eliminated at 200°C to 250°C. The remaining magnetization was of a very low intensity and was difficult to interpret. A characteristic component was

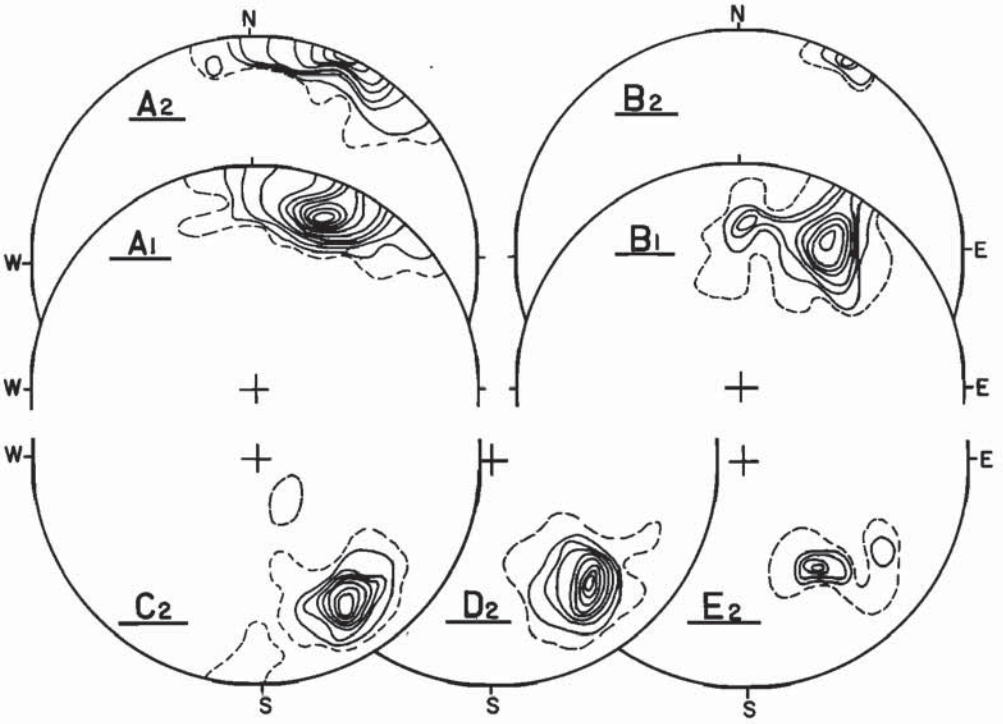


Fig. 5. Frequency distribution of early Tertiary secondary directions (corrected for bedding) in equal-area projection. A 1% counting circle was applied. Subscript 1 indicates lower hemisphere projection, and subscript 2 indicates upper hemisphere projection. Frequency intervals indicated hereunder in parenthesis : (a) Mughal Kot Fm. (1). (b) Fort Munro Fm. (1). (c) Alozai Fm. (1). (d) Brewery Limestone (1). (e) Goru Fm.-Parh Lst. at Brewery gorge (1).

eliminated in some of the specimens between 250°C and 450°C. In others, another recent field component of reversed polarity was removed in the same temperature interval (Figures 2g, 2h and 3a).

Interpretation

The pole positions, or their loci, for the bedding-corrected characteristic component directions from the various formations agree with few exceptions (Spintangi Limestone, Khadro Formation, Fort Munro Formation) with contemporaneous parts of the Indo-Pakistani APWP (Figure 6). The characteristic magnetic components are interpreted therefore as representing the primary magnetization, except for the above cited formations with aberrant results.

The main deformation of the Pakistani fold belt occurred during the Plio-Pleistocene, so the fold test is not a very decisive tool in establishing the age of acquisition of the various magnetic components. For reason of accessibility, most of our samples were collected from monoclinal sections only. In the Sanjawi area we were able to sample sections on both limbs of an anticlinal structure (i.e., Middle Jurassic and Cetaceous rocks on the northern limb and early Tertiary rocks on the Southern limb). Though not of similar age, the various characteristic directions clearly unfold in a positive sense.

Disagreement between bedding-corrected characteristic directions of the slumped top part of the nonslumped basal part of the Goru Formation-Parh Limestone section sampled at Sanjawi indicates a primary origin of the characteristic component for these formations.

The pole position of the Alozai Formation, when compared with the Indo-Pakistani APWP, falls on the Palaeozoic side of the Permo-Triassic group of poles (Figures 6a and 6d). This supports the suggested Permo-Carboniferous age (Hunting Survey, 1960) for the sampled part of this formation.

The low spread in palaeomagnetic directions obtained from the two successions of the Goru Formation-Parh Limestone at Brewery gorge and at Sanjawi, and the agreement of their pole positions with the pole for the Rajmahal Traps (100-105 m.y. Albian) (McDougall and McElhinny, 1970) indicate that the Indian plate underwent only a minor northward movement during the Aptian-Albian to Santonian-Campanian time interval represented by the sampled successions (Figures 6d and 7). The consistent normal polarity directions found throughout the sampled succession agree with the duration of the Cretaceous normal polarity interval (Irving and Pullaiah, 1976). This interval has now been dated as late Aptian to early Campanian on basis of detailed magnetostratigraphic work on the Umbrian sequence of the Apennines (Lowrie and Alvarez, 1977; VandenBerg et al., 1978; Channell et al., 1979). Seafloor spreading data from the Indian Ocean (Johnson et al., 1976; Powell, 1979) show that a phase of very rapid spreading and northward movement of the Indian plate started during the Santonian (80 m.y. B.P.) and continued until about the Palaeocene-Eocene

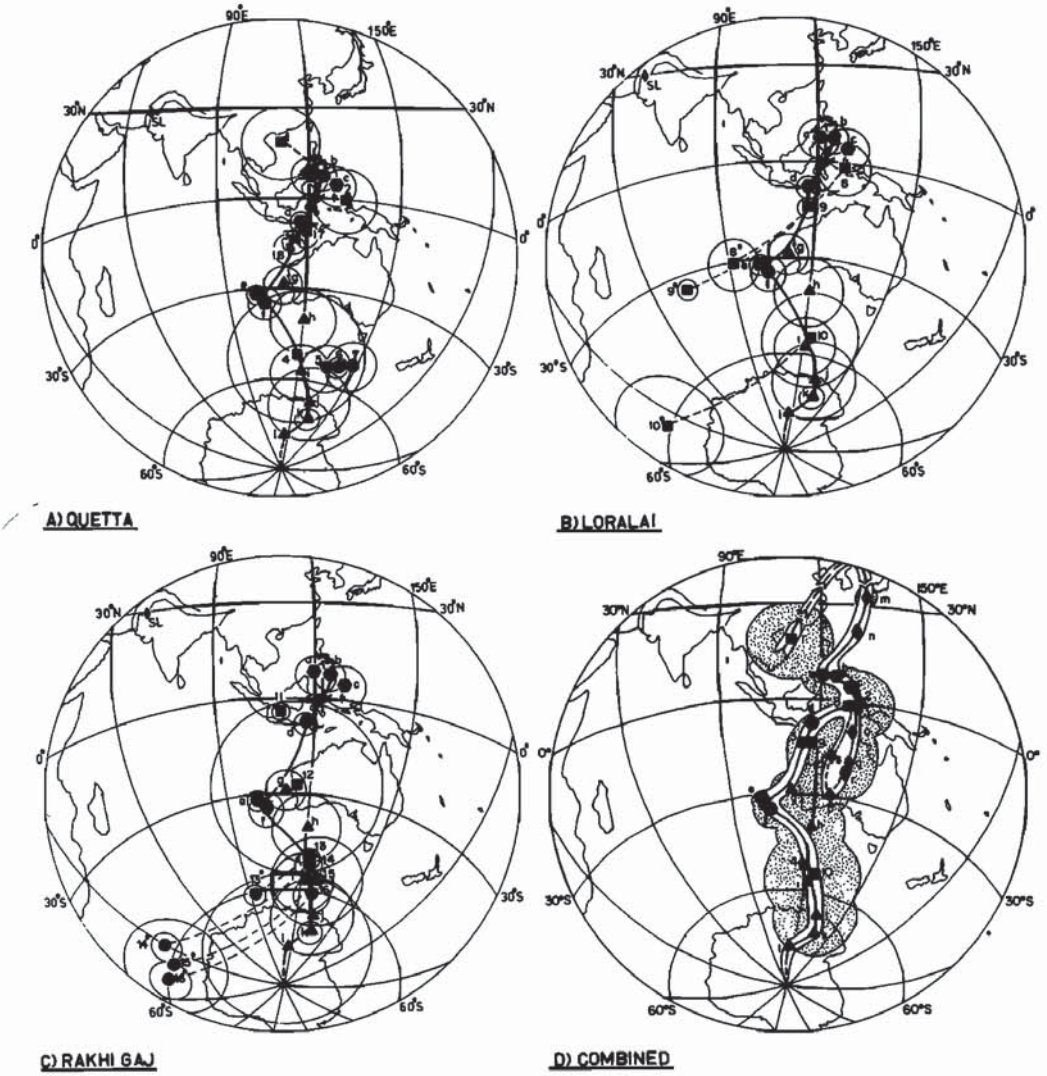


Fig. 6. Comparison of palaeomagnetic pole positions presented in this paper (squares, primary magnetizations; dots, secondary magnetization) with the (a,b,c) Indo-Pakistani APWP proposed by Klootwijk (1979a), based on selected palaeomagnetic data from the Indo-Pakistani subcontinent (hexagons) and DSDP core results from the Indian plate (triangles). Ellipses of confidence at the 95% level are also shown. (b,c) Individual pole positions have been corrected tentatively for rotation of the areas concerned. The Indo-Pakistani APWP is tentatively updated in Figure 6d on basis of (1) the primary results presented here for the Quetta area (Figure 6a) and the Loralai area (Figure 6b, corrected for rotation), (2) a renewed interpretation of existing data from the Indo-Pakistani subcontinent (diamonds m-p), and (3) Jurassic mean pole positions from Australia and Antarctica transferred to the Indian plate (diamonds r-t) according to seafloor spreading data from the Indian Ocean (see text and Table 2). Linear polar projection. Legend : 1, Alozai Fm.; 2, Chiltan Limestone; 3, Goru Fm.-Parh Lst. combined (Brewery gorge); 4, Brewery Limestone (top); 5, Alozai Fm. secondary component; 6, Brewery Limestone (base) secondary component; 7, Goru Fm. Brewery gorge secondary component; 8, Loralai Limestone; 9, Goru Fm.-Parh Lst. combined (Sanjawi); 10, Sanjawi Limestone; 11, Mughal Kot Fm.; 12, Fort Munro Fm.; 13, Khadro Fm.; 14, Mughal Kot Fm. secondary component; 15, Habib Rahi Limestone primary component; 16, Fort Munro Fm. secondary component; 17, Parh Limestone (Brewery gorge); 18, Goru Formation (Brewery gorge). For data from the Indian plate, see review by Klootwijk (1979a) : a, Panchet clays (Permo-Triassic); b, Mangli beds (Permo-Triassic Klootwijk, 1975); c, Kamthi beds, Wardha Valley (Permo-Triassic); d, Rajmahal Traps (100-105 m.y.); e, lower reversed Deccan Trap Epoch (60-65 m.y.); f, upper normal Deccan Trap Epoch (60-65 m.y.); g, DSDP cores, 75-65 m.y.; h, DSDP cores, 35-25 m.y.; 1, DSDP cores, 22.5-5 m.y. m, Talchir beds (Permo-Carboniferous); n, Kamthi beds, Wardha Valley (Permo-Triassic (Wensink, 1968); o, Pachmarhi beds (Upper Triassic); p, Parsora beds (Upper Triassic); q, mean Early Jurassic pole position according to transferred Australian data; r, mean Middle Jurassic pole position according to transferred Antarctic data; s, mean Middle Jurassic pole position according to transferred Australian data; SL, sample locality.

TABLE 2 AUSTRALIAN AND ANTARCTIC S. POLE POSITIONS TRANSFERRED TO THE INDIAN PLATE

POLE	AGE	ORIGINAL Lat.	ORIGINAL Long.	α_{95}	TRANSFERRED 1) Lat.	TRANSFERRED 1) Long.	REFERENCES
A) AUSTRALIA							
1 INTRUSIVES N.S.W.	168-181 M.Y.	51°S	186°E	11°	24°S	128°E	SCHMIDT (1976)
2 TASMANIAN DOLERITES	170 M.Y.	52°S	156°E	7°	8°S	119°E	"
3 TASMANIAN DOLERITES	170 M.Y.	51°S	175°E	5°	18°S	126°E	SCHMIDT & MCDOUGALL (1977)
4 GARRAWILLA VOLC.	193 M.Y.	46°S	175°E	10°	16°S	131°E	SCHMIDT (1976)
5 WEST. VICTORIAN BASALTS	191 M.Y.	47°S	178°E	18°	18°S	131°E	SCHMIDT (1976)
B) ANTARCTICA							
6 DOLERITIC INTRUSIONS THERON MOUNTAINS AND WHICHAWAY NUNATAKS	M.JUR.	54°S	224°E	16°	27°S	128°E	BLUNDELL & STEPHENSON (1959)
7 FERRAR DOLERITES	M.JUR. 146-162 M.Y.	58°S	218°E	5°	21°S	127°E	TURNBULL (1959)
8 FERRAR DOLERITES	M.JUR.	45°S	220°E	4°	29°S	138°E	BULL ET AL. (1962)
9 FERRAR DOLERITES	M.JUR.	59°S	221°E	18°	22°S	125°E	BRIDEN & OLIVER (1963)
10 DUFEK INTRUSIVES	168 M.Y.	57°S	192°E	7°	9°S	133°E	BECK (1972)

C) TRANSFERRED MEAN S.POLE POSITIONS

POLES	AGE	Lat.	Long.	k	α_{95}	N
AUSTRALIA, 1-3	M.JUR.	17°S	124.5°E	74	14.5°	3
AUSTRALIA, 4.5	E.JUR.	17°S	130.5°E	-	-	2
ANTARCTICA, 6-10	M.JUR.	21.5°S	130.5°E	76.5	9°	5

Rotation poles used for the India-Antarctica motion were according to Johnson et al. (1976). The rotation pole used for the Australia-Antarctica motion was according to Norton and Molnar (1977) (see Klootwyk & Peirce, 1979).

time boundary (55 m.y. B.P.). An earlier phase of spreading (Johnson et al., 1976) which resulted in about 1000 km of separation between Indo-Pakistan and Antarctica during the Aptian to Campanian interval is not recognizable in the palaeomagnetic records from the Goru Formation and the Parh Limestone. Comparison, however, of their mean result with that from the Rajmahal Traps of Albian age suggests possibly up to 500 km of northward movement of the Indo-Pakistani subcontinent during the Aptian-Albian to Santonian-Campanian time interval (Figure 7a). It is therefore likely that the early breakup of eastern Gondwanaland was characterized more by southward movement of the Antarctica-Australia block than by northward movement of the Indo-Pakistani subcontinent.

The magnetization of the sampled beds of the Khadro Formation appears to be substantially younger than its Danian Formation age. The Khadro Formation is correlated with the lower Palaeocene Cardita-Beaumonti beds of Sind (Hunting Survey, 1960). These contain lava flows which have been interpreted as the earlier phases of the Deccan Trap extrusions of the Indian mainland. The pole position of the Khadro Formation is, however, far off the Deccan Traps poles (Figure 6d). Comparison of the Khadro Formation pole, Deep-Sea Drilling Project (DSDP) core poles, and the early Tertiary poles from the Brewery Limestone and the Sanjawi Limestone (Figure 6) suggests that the Khadro Formation was remagnetized during the late Palaeocene, possibly as a result of deep weathering during the Palaeocene uplift. Ironically, this stable, well-defined secondary component is from the sole red bed unit amongst the other studied carbonate formations which have retained in part primary magnetizations.

Interpretation of the characteristic component of the Spintangi Limestone is not clear. Its bedding-corrected pole position does not conform with the early Tertiary part of the Indo-Pakistani APWP (Table 1), to which a primary component should belong, but shows affinity with the late Mesozoic pole positions instead. Neither is a secondary postfolding origin acceptable, and this result had to be rejected.

The pole positions, or the loci of the same, corresponding with the bedding-corrected secondary directions observed in the Quetta area and the Rakhi Gaj gorge, all intercept with the late Palaeocene to early Eocene part of the Indo-Pakistani APWP (Figures 6a and 6c). These secondary directions, however, cannot be interpreted in a meaningful way before bedding correction. This indicates that these secondary components were acquired during late Palaeocene to early Eocene times. This magnetic overprint may be related to initial collision of the Indo-Pakistani subcontinent with an island arc off south central Asia (Klootwijk 1979a,b; Klootwijk et al., 1979). A comparison of palaeomagnetic results from the Ladakh Intrusives north of the Indus-Tsangpo suture in the NW Himalaya (Klootwijk et al., 1979) with DSDP core data from the Indian plate indicates that this initial collision had occur-

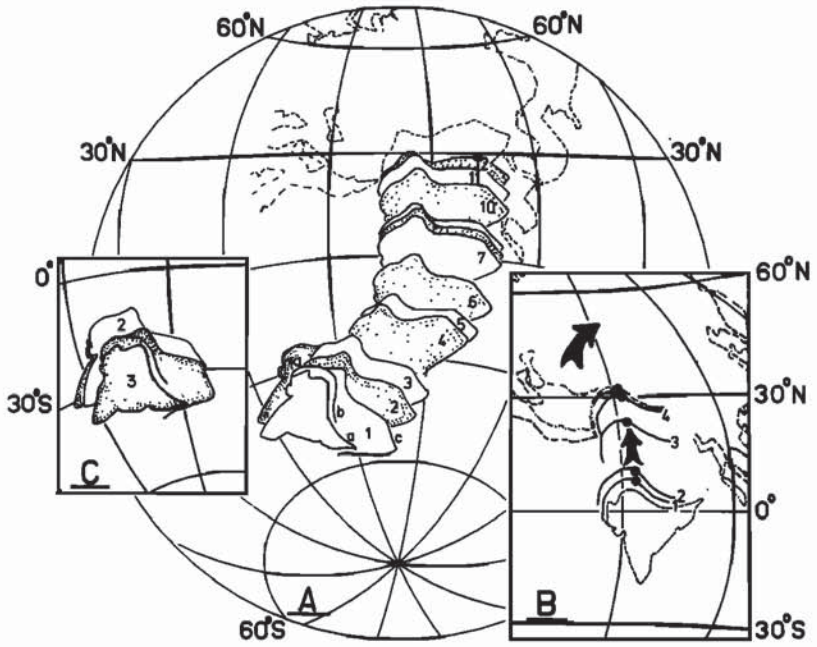


Fig. 7. (a) Successive latitudinal orientations of Greater India (Veevers et al., 1975) according to the palaeomagnetic data presented here and date summarized by Klootwijk (1979a). The longitudinal positions cannot be determined by the palaeomagnetic method used and are arbitrary. Linear polar projection. Legend : 1, mean Permo-Triassic result (Kamthi, Mangli, and Panchet beds); 1a, Loralai Limestone, Middle Jurassic; 2, Rajmahal Traps, 100-105 m.y.; 3, Goru Fm.-Parh Lst. (Brewery gorge) Aptian-Albian to Santonian-Campanian; 4, lower reversed Deccan Trap epoch, 60-65 m.y.; 5, upper normal Deccan Trap epoch, 60-65 m.y.; 6, DSDP cores, 65-55 m.y.; 7, Khadro Fm., late Palaeocene?.; 8, Sanjawi Limestone, late Palaeocene-early Eocene; 9, DSDP cores, 55-45 m.y.; 10, DSDP cores, 45-35 m.y., 11, DSDP cores 35-25 m.y.; 12, DSDP cores 22.5-5 m.y. This sequence suggests a rather gradual northward and counterclockwise rotational movement during late Mesozoic and Cenozoic times, preceded during the early Mesozoic by a more complex pattern of movements (1a, see also Figure 8c). The three zones drawn at the northern periphery of the Indo-Pakistani subcontinent (1) represent from south to north the following : a, the Baluchistan-Himalayan-Arakan Yoma front; b, the Indus-Tsangpo suture zone; c, the northern boundary of Greater India according to Veevers et al. (1975). For comparison, part of the present outline of Eurasia also is relocated according to Irving (1977); dashed-dotted lines is Permo-Triassic position, and dashed line is position at about 20 m.y. B.P. (b) Successive palaeolatitudinal positions of the Ladakh Intrusives at Kargil (Klootwijk et al., 1979) according to various magnetic components, compared with southern Eurasia's position (Irving, 1977) at about 50 m.y. ago (dashed line) and at about 20 m.y. ago (dashed-dotted line). The dots represent the sampled locality of the Ladakh Intrusives; 1 and 2, positions upon cooling of the batholith (about 50 m.y. ago) after initial collision of the Indo-Pakistani subcontinent with an island arc off south central Asia; 3, and 4, positions possibly associated with much younger orogenic phases (Miocene and younger). To enhance visibility, only the present outline of the Indus-Tsangpo suture zone (solid line) and the outline of the Indo-Pakistani subcontinent (dashed line according to component 1) are shown in their present form and position in relation to the Kargil locality. Aitoff projection. (c) Middle Mesozoic palaeolatitudinal positions for the Indo-Pakistani subcontinent, based on Australian and Antarctic pole positions transferred to the Indian plate (Table 2). Linear polar projection: 1, mean Early Jurassic position according to transferred Australian data. 2, mean Middle Jurassic position according to transferred Antarctic data. 3, mean Middle Jurassic position according to transferred Australian data.

red already by the early Eocene (Figure 7). This timing of the initial collision agrees with the Palaeocene, possibly early Eocene, emplacement of ultramafic complexes in eastern Baluchistan (Allemann, 1979), with early Eocene detritus being spread onto the northeastern margin of the Indian shield from a granitic trans-suture source (Stoneley, 1974), and with the sudden and abundant occurrence of middle Eocene land mammals of Eurasian affinity (Sahni and Kumar, 1974) in northwestern Indo-Pakistan. Moreover, such a timing coincidences with an evident hiatus in the sedimentation record of the Indian Ocean. This hiatus may have lasted from the Palaeocene to possibly the middle Eocene (Curray and Moore, 1974; Pimm and Sclater, 1974; Davies et al., 1975).

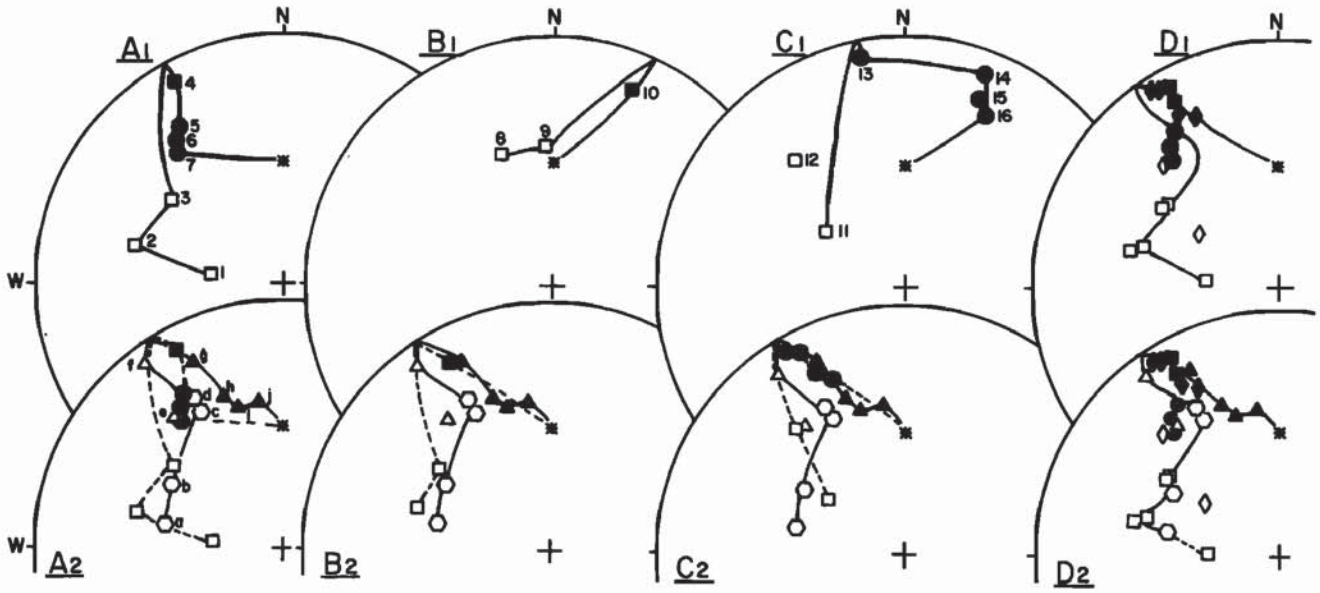


Fig. 8. (a) Palaeomagnetic mean directions (corrected for bedding) obtained in the Quetta area (squares are primary directions, dots are secondary directions, and asterisks are present field direction), compared with expected directions in the sampled area according to available palaeomagnetic data from the Indo-Pakistani subcontinent (hexagons) and DSDP core results from the Indian plate (triangles). (b) As above, for the Sanjawi area. In Figure b2, directions actually obtained have been corrected for an interpreted clockwise rotation over 50°. (c) As above, for the Rakhi Gaj gorge.In Figure c2 the Khadro Fm. result (13) has been tentatively corrected for a resultant clockwise rotation over 20°, and the secondary components (14-16) have been tentatively corrected for a 50° clockwise rotation. (d1) combined directions from the three areas after corrections for tectonic rotation, diamonds are tentatively corrected primary and secondary directions from the Rakhi Gaj gorge (see Figure 6c). (d2) As above, but now compared with expected directions at the Sanjawi area according to available palaeomagnetic data from the Indian plate. All figures are in equal-area projection. Legend : for poles 1-16, see caption to Figure 6. For poles a-j, see review by Klootwijk (1979a); a, mean Permo-Triassic result; b, Rajmahal Traps (100-105 m.y.; c, lower reversed Deccan Trap Epoch (60-65 m.y.); d, upper normal Deccan Trap Epoch (60-65 m.y.); e, DSDP cores, 75-65 m.y.; f, DSDP cores, 65-55 m.y.; g, DSDP cores, 55-45 m.y.; h, DSDP cores, 45-35 m.y.; i, DSDP cores, 35-25 m.y.; j, DSDP cores, 22.5-5 m.y.

The Palaeocene crustal upwarp of the studied region, evidenced by marine regressions in the Quetta and Rakhi Gaj gorge areas, can be related to this initial collision and may represent an early Tertiary predecessor of the present-day topographic rise of the central Indian Plateau. Warsi and Molnar (1977) suggest that this rise may be similar to lithospheric flexures presently found seaward of oceanic trenches.

Acquisition and stabilization of these secondary components may be attributed to cooling upon uplift (Pullaiah et al., 1975), possibly enhanced by increased heat flux near the collision zone. Absence of secondary components in the Sanjawi area is to be expected, since the Sanjawi Arch already formed a topographic high during Cretaceous and Palaeocene times (Hunting Survey, 1960), prior to initial collision.

Implications for Postcollision Tectonics

The primary and secondary magnetic directions show consistent and appreciable differences between the various areas studied, which can be interpreted in terms of major tectonic rotations with respect to the Indian shield. The agreement between expected and observed declinations of the primary components in the northern extreme of the Kirthar Range indicates absence of significant rotation of that part of the Kirthar Range (Figures 6a, 8a1, and 8a2 and Table 1) with respect to the Indian shield. The slight disagreement between expected and observed directions for the secondary components needs, however, further testing.

Declinations of the Middle Jurassic to Palaeocene-Eocene results from the Sanjawi area (central Loralai Range) are consistently deflected from the expected Indian shield pattern (Figures 6b, 7b1, and 7b2 and Table 1) and indicate a post-early Eocene clockwise rotation over 50° with respect to the Indian shield.

The palaeomagnetic directional pattern from the Rakhi Gaj gorge in the southern Sulaiman Range is more complex (Figures 6c,8c1 and 8c2 and Table 1) than that from the two other areas and cannot be interpreted as yet in a meaningful way. The precisely determined result from the Mughal Kot Formation does not fall on the expected Campanian-Maastrichtian segment of the Indo-Pakistani APWP, but the less precise Fort Munro Formation result (Maastrichtian) agrees rather

well with this APWP. Together, both late Cretaceous results suggest no significant rotation with respect to the Indian shield. In contrast, the result from the Khadro Formation (Danian formation age, but a probable late Palaeocene magnetic age, as will be discussed) is suggestive of a clockwise rotation of the sampled area over 20°, whereas the three early Tertiary secondary components suggest a clockwise rotation over 50°. This sequence of palaeomagentic results is puzzling and cannot be explained by our current knowledge about the tectonics of the region. The palaeomagnetic data could be reconciled tentatively by assuming a two-stage rotational movement model. Such an ad hoc explanation would invoke a first phase of counterclockwise rotation which started after the Maastrichtian and which should have proceeded over 30° upon acquisition of the characteristic component of the Khadro Formation (probably late Palaeocene). At the time of acquisition of the late Palaeocene to early Eocene magnetic components this counterclockwise rotation should have accumulated over at least 50°. In a subsequent phase this movement should have reversed to a clockwise rotation which to date proceeded up to 50°.

There are, however, no tectonic indications for the hypothized early Tertiary counterclockwise movement. The Sulaiman Range may have rotated during the major Plio-Pleistocene tectonic phase, but the regional tectonics suggest a counterclockwise rather than a clockwise movement. It is more likely that the puzzling pattern of primary and secondary directions has resulted from local distortions associated with a recently recognized major NNW-SSE directed sinistral shear zone (Kingri fault) which passes very near to the sampled locality (Andrieux and Brunel, 1977; Rowlands, 1978; Kazmi, 1979; C. McA. Powell personnal communication, 1978). Further palaeomagnetic work is needed to resolve this palaeomagnetic-tectonic paradox.

The palaeomagnetic observation of an evident clockwise rotation of the Loralai Range supports arguments for an allochthonous nature of the Sulaiman lobe (Templeton, 1973; Powell, 1979; Sarwar and De Jong, 1979). High seismic activity along the southern front of the Sulaiman lobe (Menke and Jacob, 1976; Quittmeyer et al., 1979; Seeber and Armbruster, 1979) and along the eastern boundary of the Sulaiman Range (Nowroozi, 1972; Menke and Jacob, 1976; Rowlands, 1978; Seeber and Armbruster, 1979; Quittmeyer et al., 1979; L. Seeber personal communication, 1979) indicates ongoing movement of the Sulaiman lobe towards the Indian shield.

It is open to further study whether the whole zone between the Loralai-Sulaiman festoon and the Chaman fault zone (the region comprising the fold belts, the ultramafic bodies in the Zhob Valley, and the Tertiary flysch formations from the Zhob-Kakar region) has undergone a clockwise rotation, since movements of the more interior parts of the lobe remain to be established. Correction of the Sulaiman lobe for a hypothetical 50° clockwise rotation around a pole near Quetta relocates the Zhob Valley ultramafics near the Chaman fault zone (which can be interpreted as the original zone of obduction of these ultramafic bodies) and relocates the Zhob-Kakar flysch at the Makran side of the Chaman fault zone. A further correction for late Tertiary northward movement of these flysch formations, while positioned on the northward moving Indo-Pakistani subcontinent, relocates the Zhob-Kakar flysch much nearer to the northern Makran flysch. The Zhob-Kakar flysch thus may be part of the arc-trend sedimentary sequence deposited on the oceanic frontal part of the Afro-Arabian plate, which has been subducting underneath the Makran-Ras Koh-Chagai region since the Late Cretaceous (Farhoudi and Karig, 1977; Quittmeyer et al., 1979; Jacob and Quittmeyer, 1979; White, 1979; Arthurton et al., 1979).

Indo-Pakistan's APWP and its Northward Drift

The primary palaeomagnetic poles from the northern Kirthar Range and the rotation-corrected poles from the Loralai Range are consistent with the pole positions from the Indian plate (Figures 6a and 6b). The Indo-Pakistani APWP has been updated in accordance with these new results (Figure 6d), and various stages in the northward drift of the Indo-Pakistani subcontinent are shown in Figure 7. There are two major kinks in this updated APWP (Figure 6d); one is defined by the middle and late Jurassic poles from the Chiltan Limestone and the Loralai Limestone, and the other is defined by the early Tertiary Deccan Trap poles. The Jurassic kink may be associated with breakup of Gondwanaland and the early rifting of the Indian plate away from the Australia-Antarctica plate (Markl, 1974; Larson, 1975, 1977; Larson et al., 1978). The shape of this kink has to be studied further. This kink may be more extensive and in fact may form part of a Triassic-Jurassic loop. This loop is tentatively drawn in Figure 6d according to Jurassic mean pole positions from the Australia-Antarctica plate transferred to the Indian plate (Figures 6d and 7d and Table 1) and late Triassic pole positions from Indian Gondwana red beds. Recently obtained early Mesozoic palaeomagnetic results from the Tibetan Sedimentary Series of the Nepal Himalaya (Klootwijk and Bingham, 1980) and a reappraisal of lower Gondwana red beds from north eastern India (V.K. Agarwal and C.T. Klootwijk, manuscript in preparation, 1980) strongly support the reality of such a loop. The early Tertiary kink coincides with and follows a period of very fast spreading in the Indian Ocean (McKenzie and Sclater, 1971 : 75-55 m.y. ; Johnson et al., 1976, and Powell, 1979 : 80-53 m.y.) and may reflect a change in the pattern of plate movements.

The primary and secondary palaeomagnetic directions of early Tertiary age obtained here support the conclusion drawn from palaeomagnetic observations in Ladakh and the eastern Hindu Kush (Klootwijk et al., 1979; Klootwijk, 1979b; Klootwijk and Conaghan, 1979) that the late Palaeocene-early Eocene collision of continental Indo-Pakistan with the Kohistan-Ladakh island arc off south central Asia occurred at very low northern palaeolatitudes (Figures 7a and 7b). Subsequent northward movement of this combined unit over about 3000 km resulted in large-scale indentation of south central Asia.

Concluding Remarks

The palaeomagnetic results presented in this study may illucidate the tectonic evolution of

the northwestern part of the Indo-Pakistani subcontinent. It is well known from the geological record that in late Mesozoic times a thick sedimentary pile accumulated between two northwest aligned basement highs, the Sarghoda high trending toward the Bannu reentrant in the north and in the south the Jacobabad high with its northwestern continuation into the Sibi Trough (Hunting Survey, 1960; Voskresenskiy et al., 1968; Auden, 1974). During the Palaeocene the ultramafic complexes of the Zhob Valley and Waziristan were emplaced. The Sulaiman lobe moved south mainly during the late Tertiary by a supracrustal décollement over the counterclockwise rotating and northward translating Indo-Pakistani subcontinent. This décollement was facilitated possibly by Infra-Cambrian evaporites similar to the Infra-Cambrian Salt Range Formation. Thick evaporitic sequences are present in the subsurface near Multan about 160 km east of the Sulaiman Range, and continuation of the same facies underneath the Sulaiman lobe is likely.

This clockwise rotation of the Loralai Range, strongly suggested by the palaeomagnetic data presented here, is directly related to the post-Eocene large-scale indentation of the Indo-Pakistani subcontinent into south central Asia, which ultimately decoupled the Sulaiman lobe from the Indian shield. The festoonlike shape of this lobe and the hairpinlike Quetta Syntaxis probably resulted from obstruction of the allochtonous Sulaiman lobe in the region near Quetta, possibly because of pinching out of the evaporite facies but more probably through obstruction by the thick molasse wedge of the Sibi trough.

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